

Overview

In this assignment, Starter Project #3, we were tasked with taking our Boggle Solver Code and submitting it to a partner code review. We had to push our original solver code to GitHub and invite our partner to review it, while we reviewed our partner's. During the code review, we made in-line comments on what could be improved in our partner's Boggle Solver code, and made an overall comment that summarizes our comments and corrections. After we were done reviewing each other's codes, we then made changes to our own to correct what was pointed out during the review. Then, we looked toward adjusting our final and corrected code to a format called PEP 8. While adjusting the code, we conducted regression tests through the Codio interface to ensure we did not affect the functionality of the code. Once both the Boggle Solver code and Boggle Test code passed the Codio auto-grader, we were then asked to summarize our experience during the entire process in a report.

Feedback Received

During the review, I received three in-line comments and one overall comment. In my in-line comments, my partner suggested that I remove my `setGrid` and `setDictionary` functions because I didn't actually use them in my solver, "This may be unnecessary. I don't see the method used anywhere in the code." My reviewer also gave positive feedback on my `dict_hash` function, stating, "great function. function is building a prefix hash to stop unnecessary searches early". The overall comment that I received from my partner was, "Great program. Overall the program is very easy to read and follow along, especially with the organized and consistent comments." I agreed with all of the corrections my partner suggested and appreciated the positive feedback I got as well.

Feedback Given

On my end, I made in-line comments that pertained to improving the readability of my partner's code, like "Really great function! We could improve readability if the comments are aligned with the line it is referring to (i.e. the indentation is the same for both the comment and line of code it explains)." Also, I noticed some debugging code still present in the final solver code, so I suggested that those lines be removed: "I love that you included debugging code (`print(':')`), `print('2')`, `print(found)`) during your process! Now that we have the final solution code, removing it would clean up the output of the program!" To go along with that, I left a positive review that spoke on my partner's method of marking already visited cells in her solver, "Great implementation of backtracking logic to mark visited cells!" Lastly, my overall review comment

on my partner's solver code was "Overall great code! Your use of comments and white space really aided in the understanding and readability of your code. I think your approach to the solution was done very well!"

Improvements Implemented

As far as improvements I implemented on my Boggle Solver Code, I agreed with my reviewer's comments about removing the unused functions `setGrid` and `setDictionary`. In that regard, I removed both functions, checked the code for functionality, and then pushed the finalized code to GitHub.

Static Analysis Results

Another part of this assignment involved adjusting both our Boggle Solver and Boggle Test codes to fit the PEP 8 format. While I had not heard of this type of format before, I read the provided materials on what was expected and fixed my code as necessary. In both my solver and test codes, I had to fix issues like trailing whitespace, exceeding character limit in one line, blank line containing whitespace, newline at the end of file, two blank lines between classes and functions, and switching from two space to four space indentations.

Regression Testing

During the process of correcting and adjusting my codes, I made sure to make use of the embedded test suite in codio. Each time a change was made, I ran the test to see if it passed or not. When it came to the solver, if the test suite was passed for the PEP 8 format, then I would run the auto-grader to make sure the solver was still functional. There was some back and forth between the PEP 8 check and the auto-grader, but when both passed, I concluded my revisions.

Reflection

From this assignment, I learned a lot about code reviews and professional workflows. I was exposed to how to request a review on my own branch in a repository, and how to complete reviews requested by other people. There was a lot of insight given on how to give comments in a professional and respectful manner. Also, how important it is to give both suggestions on improvement and positive comments that acknowledge what was done well. I was also introduced to the format style of PEP 8 for the first time, which allowed me to practice a new style and practice regression testing. Overall, Starter Project #3 was a great learning experience

that exposed me to new concepts that I was able to put into action and practice so that I will be better prepared when entering a professional environment where these things are routine.